

Mizanur Rahman

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 [developerhridu](#) |  [developerhridu](#) |  [Portfolio](#)

RESEARCH INTERESTS

Machine Learning Systems, Database Systems, In-Database ML and SQL/ML Co-Optimization, Applied Deep Learning & Computer Vision, and Spatiotemporal / Geospatial Data Systems.

SUMMARY

Software engineer with **4+ years** building secure, fault-tolerant backend and database systems for high-availability, transaction-heavy platforms, now seeking an **MS in Computer Science** to pursue research at the intersection of **database systems and machine learning**. Combines production-grade systems engineering — **PostgreSQL migration, CQRS/ETL pipelines**, query optimization, and distributed microservices — with hands-on **deep learning** experience from a **CNN/VGG computer-vision thesis**. Motivated to contribute to systems that co-optimize databases and ML, including in-database machine learning and geospatial AI.

EDUCATION

Hajee Mohammad Danesh Science & Technology University

BSc (Eng.) in Electronics & Communication Engineering (CGPA – 3.1/4.0) [Transcript]

Jan. 2017 – Jan. 2023

Dinajpur, Bangladesh

RESEARCH EXPERIENCE

Undergraduate Thesis — Deep Learning for Produce Freshness Detection

2022 – 2023

Supervisor: Prof. Dr. Md. Mahabub Hossain; Co-supervisor: Asst. Prof. Jannatun Ferdous

HSTU, Dinajpur

- Designed and trained a custom **deep CNN (VGG-style)** to detect and classify fruit & vegetable freshness across **24 classes**, reaching **~98% test accuracy** and **outperforming AlexNet, VGG-16/19, and ShuffleNetV2** baselines on both freshness and fruit-classification tasks.
- Built the end-to-end pipeline: assembled a **9,000+ image** dataset (web scraping + Kaggle) and implemented preprocessing (denoising, auto-cropping, resizing to 224×224) and **data augmentation** to mitigate overfitting on a small dataset.
- Ran systematic ablations across **color spaces** (Grayscale/RGB/HSV) and **kernel sizes**; best configuration (HSV + grayscale + hue/saturation + flips) achieved **96.4%** test accuracy.
- Trained on **Google Colab/Cloud** (Tesla K80 GPU) using **Keras/TensorFlow**, applying batch normalization, dropout, ELU/ReLU activations, and scheduled learning-rate decay.
- Analyzed **model-size / edge-deployment** constraints and proposed **knowledge distillation** and region-specific fine-tuning as future work — a systems-aware view of ML deployment.

TECHNICAL SKILLS

Machine Learning : Python, TensorFlow, Keras, CNN/VGG Architectures, Computer Vision, Data Augmentation, Model Evaluation

Data Analytics : Pandas, PySpark, GeoPandas, Matplotlib, and Plotly

Databases : MSSQL Server, PostgreSQL, MongoDB, Redis; Query Optimization, Read/Write Separation, DB Migration

Backend : C#, .NET Core, ASP.NET MVC, EF Core, Dapper, LINQ, API Integration, REST APIs, SOAP

Architecture : Microservices, Clean Architecture, CQRS, Multi-tenant SaaS, Event-Driven Design, ETL Pipelines

Messaging & Caching: RabbitMQ, Kafka, Redis, Hangfire

Frontend : Angular, TypeScript, JavaScript, React, HTML/CSS, Bootstrap

DevOps & Observability: Docker, Git, GitHub, GitHub Actions, GitLab CI/CD, APM, Elasticsearch

Testing : xUnit, MSTest, Jasmine, Postman, JMeter, k6

PROFESSIONAL EXPERIENCE

TechnoNext Software Ltd. (Concern of US-Bangla Airline)

Software Engineer

Apr. 2024 – Present

Dhaka, Bangladesh

Online Travel Agency: A multi-tenant OTA platform that enables travel agencies to search, compare, book, and manage flights, hotels, holiday packages, and other travel services.

Tech-Stack: Microservices, .NET Core, Dapper, Angular, React, MSSQL Server, MongoDB, RabbitMQ

- Built scalable software solutions for **6 OTA clients** including TripLover, TravelChamp, FirstTrip, TakeTrip, TakeOff, & TripLover Dubai.
- Contributed to the OTA dynamic ecosystem comprising **36+ services**, ensuring scalability, high availability, and fault tolerance.
- Architected and led development of a **multi-tenant OTA SaaS** project supporting both **B2B & B2C** business models, enabling a single codebase to serve multiple clients with isolated configurations.
- Built an automated **API integration test suite** using **xUnit** for **30+ supplier integration projects**, ensuring consistent API behavior and accelerating release validation.
- Led the **migration** of the core database from **SQL Server to PostgreSQL**, reducing licensing costs and improving cross-platform compatibility on Linux infrastructure.
- Integrated **APM** monitoring with **Elasticsearch**-based structured **logging**, cutting production failure rates by **30%** and dramatically reducing mean-time-to-debug.
- Developed a background service using **ASP.NET Core & Hangfire** that automatically cancels un-ticketed bookings before the last ticketing time, reducing ADM penalties by **\$4,500–\$6,400/month** per OTA and saving approximately **\$23,000–\$27,000** overall across 6 clients.
- Implemented key OTA features — **Progressive Search, Search Ranking, Universal Split PNR, Refund Reissue, PNR Details**.
- Delivered cloud-native **8** supplier integration projects (**Sabre, Travelport, Travelport NDC, Air India, Flydubai, SQ NDC, TK NDC, & Indigo**), increasing flight search feed by **27%**.
- Migrated **8** legacy .NET Framework applications to .NET Core, enabling **cross-platform** deployment on **Linux servers** and reducing infrastructure costs by eliminating Windows Server licensing.
- Contributed to **CI/CD pipelines** and **code reviews**, and supported **onboarding** & mentoring of new team members.

Foodi : A food delivery platform that enables customers to browse restaurants, place orders, make payments, and track deliveries in real time.

Tech-Stack: Microservices, .NET Core, EF Core, Go, Node.js, PostgreSQL, React, RabbitMQ, Kafka

- Improved API response time by implementing a **read/write Redis caching** strategy for better scalability.
- Designed and developed the **Engagement Automation** feature for Foodi-CRM to streamline customer & operation team communication workflows.

Fare Comparison: An airline fare intelligence platform that enables airlines to compare competitors' real-time fares, flight schedules, and seat availability, providing data-driven pricing recommendations to maximize revenue and profitability.

Tech-Stack: .NET Core, Hangfire, EF Core, React, PostgreSQL

- Designed and developed the end-to-end platform, delivering a scalable and responsive web application.
- Integrated multiple GDS, NDC Suppliers to aggregate competitor fares, flight schedules, and seat availability, normalizing data for accurate comparison and analysis.
- Engineered pricing-intelligence logic to analyze market data and generate data-driven fare recommendations, helping airlines optimize ticket pricing and maximize revenue.

Bdjobs.com Ltd.

Jr. Software Engineer [Verify]

May 2023 – Mar. 2024

Kawran Bazar, Dhaka

Bdjobs.com & Recruiter.Bdjobs.com : Platforms that connect job seekers with employers by enabling job posting, job search, application management, resume submission, and recruitment workflows.

- Implemented a **CQRS-style architecture** with an **ETL pipeline** separating read/write databases for job listings; **database query optimization** reduced query time and improved search efficiency by **30%**.
- Collaborated with **product, backend & QA teams** to migrate Bdjobs.com corporate panel from legacy ASP to Angular and ASP.NET Core, enhancing platform functionality and performance.

- Developed key features — **Employers Job Posting and Hiring System** and **Candidate SkillMatch Score**, increasing platform sales by **17%**.

E-Desh Limited

May 2022 – Feb. 2023

Jr. Software Engineer (Part-Time) [[Proof](#)]

Remote

Courier Platform: A comprehensive logistics platform that streamlines parcel booking, real-time tracking, last-mile delivery, reverse logistics, fulfillment, and cross-border shipping through a centralized management system.

- Collaborated with a team of remote developers to design and develop the company's website using **React** and **ASP.NET Core**.
- Crafted front-end (React), built APIs, and optimized existing **codebase & database**, resulting in improved website performance and responsiveness.

PROJECTS

Documentation Platform

[Back-End Code](#) & [Front-End Code](#)

- A knowledge-sharing platform that enables users to create, organize, search, and access technical documentation, tutorials, API references, and best practices through a centralized repository.
- Tech-Stack: .NET Core, EF Core, Angular, MSSQL Server

Expense Tracker

[Code](#)

- A cross-platform personal finance application for **Android** and **iOS** that enables users to track cash and credit card expenses, record income, manage budgets, categorize transactions, and monitor spending through interactive reports and analytics. Tech-Stack: Flutter, Firebase

TRAINING AND CERTIFICATIONS

Cloud Architecture Training [Verify]	Bdjobs.com Ltd. & DU IIT	Jun. 2024
Web Design and Development [Verify]	ICT Division, Bangladesh (LEDP)	Feb. 2020
Certified Software Engineer [Verify]	HackerRank	Jan. 2024
Certified SQL (Advanced) [Verify]	HackerRank	Jul. 2026
Certified C# Developer [Verify]	HackerRank	Feb. 2024
Certified JavaScript Developer [Verify]	HackerRank	Jul. 2026
Rest API (Intermediate) [Verify]	HackerRank	Jul. 2026
React Basics [Verify]	Coursera (Meta)	Apr. 2023
Introduction to Back-End Development [Verify]	Coursera (Meta)	Jun. 2023
Introduction to Front-End Development [Verify]	Coursera (Meta)	Feb. 2023

REFERENCES

Md. Sumsuddin Shojib

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Dr. Md. Mahabub Hossain

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